hw0303

2026-03-04

```
library(readxl)
```

```
## Warning: 套件 'readxl' 是用 R 版本 4.2.3 來建造的
```

```
star5_small <- read_excel("C:/finance ecomatrics/data/star5_small.xlsx")
head(star5_small)
```

```
## # A tibble: 6 × 7
##    aide mathscore readscore regular small tchexper totalscore
##   <dbl>     <dbl>     <dbl>   <dbl> <dbl>    <dbl>      <dbl>
## 1     0       528       461       0     1       13        989
## 2     0       454       436       1     0        8        890
## 3     1       463       395       0     0        8        858
## 4     1       449       430       0     0        3        879
## 5     0       478       421       0     1       15        899
## 6     1       536       445       0     0       11        981
```

```
summary(star5_small)
```

```
##       aide          mathscore       readscore        regular      
##  Min.   :0.0000   Min.   :339.0   Min.   :358.0   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:454.0   1st Qu.:414.0   1st Qu.:0.0000  
##  Median :0.0000   Median :478.0   Median :433.0   Median :0.0000  
##  Mean   :0.3542   Mean   :485.9   Mean   :435.8   Mean   :0.3433  
##  3rd Qu.:1.0000   3rd Qu.:513.0   3rd Qu.:451.0   3rd Qu.:1.0000  
##  Max.   :1.0000   Max.   :626.0   Max.   :627.0   Max.   :1.0000  
##                                                                   
##      small           tchexper        totalscore    
##  Min.   :0.0000   Min.   : 0.000   Min.   : 727.0  
##  1st Qu.:0.0000   1st Qu.: 5.000   1st Qu.: 870.0  
##  Median :0.0000   Median : 9.000   Median : 915.0  
##  Mean   :0.3025   Mean   : 9.126   Mean   : 921.6  
##  3rd Qu.:1.0000   3rd Qu.:13.000   3rd Qu.: 963.0  
##  Max.   :1.0000   Max.   :27.000   Max.   :1253.0  
##                   NA's   :6
```

2.22 (a)

```
data_a <- subset(star5_small, small==1 | regular==1)
model_a <- lm(totalscore ~ small, data = data_a)
summary(model_a)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = data_a)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -189.44  -50.44   -9.44   43.06  324.38
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.675 249.396   <2e-16 ***
## small         12.175      5.369   2.268   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,   Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

Based on the R output provided, the estimated regression equation is:

$$TOTALSCORE = 916.442 + 12.175 \times SMALL$$

1. Intercept (β_1): 916.442 This represents the estimated average total score for students in regular-sized classes (where small = 0). 2. Slope for small (β_2): 12.175 On average, students assigned to small classes score 12.175 points higher than students in regular-sized classes

Conclusion: We conclude that class size has a statistically significant effect on student learning. Specifically, being in a small class positively impacts students’ academic performance, leading to higher total scores on average.

2.22(b)

```
model_b_read <- lm(readscore ~ small, data = data_a)
summary(model_b_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = data_a)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -74.665 -21.590  -2.665  16.410 194.335
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.488 290.760  < 2e-16 ***
## small          6.924      2.174   3.185  0.00151 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.2 on 773 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.01167
## F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
model_b_math <- lm(mathscore ~ small, data = data_a)
summary(model_b_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777  -35.028   -5.028   29.223  142.223
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.449 197.545   <2e-16 ***
## small          5.251      3.578   1.467    0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778,   Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

for the Reading Score Model, estimated equation:

$$READSCORE = 432.665 + 6.924 \times SMALL$$

1. Intercept (432.665): The estimated average reading score for students in regular-sized classes is 432.665.
2. Slope for small (6.924): Students in small classes score, on average, 6.924 points higher in reading than those in regular classes. The p-value of 0.00151 indicates statistical significance.

for the Math Score Model, estimated equation:

$$MATHSCORE = 483.777 + 5.251 \times SMALL$$

1. Intercept (483.777): The estimated average math score for students in regular-sized classes is 483.777.

2. Slope for small (5.251): Students in small classes score, on average, 5.251 points higher in math than those in regular classes. The p-value of 0.143 indicates no statistical significance.

Conclusion: This suggests that class size reduction may be effective in improving reading skills, but fails to provide significant improvements in quantitative and math skills.

2.22(c)

```
data_c <- subset(star5_small, aide==1 | regular==1)
model_c <- lm(totalscore ~ aide, data = data_c)
summary(model_c)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442   40.558  312.558
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.559 257.529   <2e-16 ***
## aide           4.306      4.994   0.862    0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898,  Adjusted R-squared:  -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

Based on the R output provided, the estimated regression equation is:

$$TOTALSCORE = 916.442 + 4.306 \times AIDE$$

1. Intercept (γ_1): 916.442 This represents the estimated average total score for students in regular-sized classes (where aide = 0). 2. Slope for small (γ_2): 4.306 students in regular-sized classes with a teacher aide score, on average, a mere 4.306 points higher than those in regular classes without an aide. But the p-value of 0.389 indicates no statistical significance.

Conclusion: Because the p-value is not significant, we fail to reject the null hypothesis. We conclude that adding a full-time teacher aide to a regular-sized classroom has no statistically significant effect on students' overall learning (total scores).

2.22(d)

```
model_d_read <- lm(readscore ~ aide, data = data_c)
summary(model_d_read)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = data_c)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -74.665 -21.536  -2.536  15.464 194.335
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.460 296.341   <2e-16 ***
## aide           2.871      2.049   1.401    0.161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.64 on 835 degrees of freedom
## Multiple R-squared:  0.002347,   Adjusted R-squared:  0.001152
## F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
model_d_math <- lm(mathscore ~ aide, data = data_c)
summary(model_d_math)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -146.212  -31.212   -5.777   29.223  142.223
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.355 205.464   <2e-16 ***
## aide           1.435      3.304   0.434    0.664
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.79 on 835 degrees of freedom
## Multiple R-squared:  0.0002258,  Adjusted R-squared:  -0.0009715
## F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

for the Reading Score Model, estimated equation:

$$READSCORE = 432.665 + 2.871 \times AIDE$$

1. Intercept (432.665): The estimated average reading score for students in regular-sized classes without an aide is 432.665.
2. Slope for aide (2.871): Having a teacher aide increases the reading score by an average of 2.871 points. However, with a p-value of 0.161, this effect is not statistically significant.

for the Math Score Model, estimated equation:

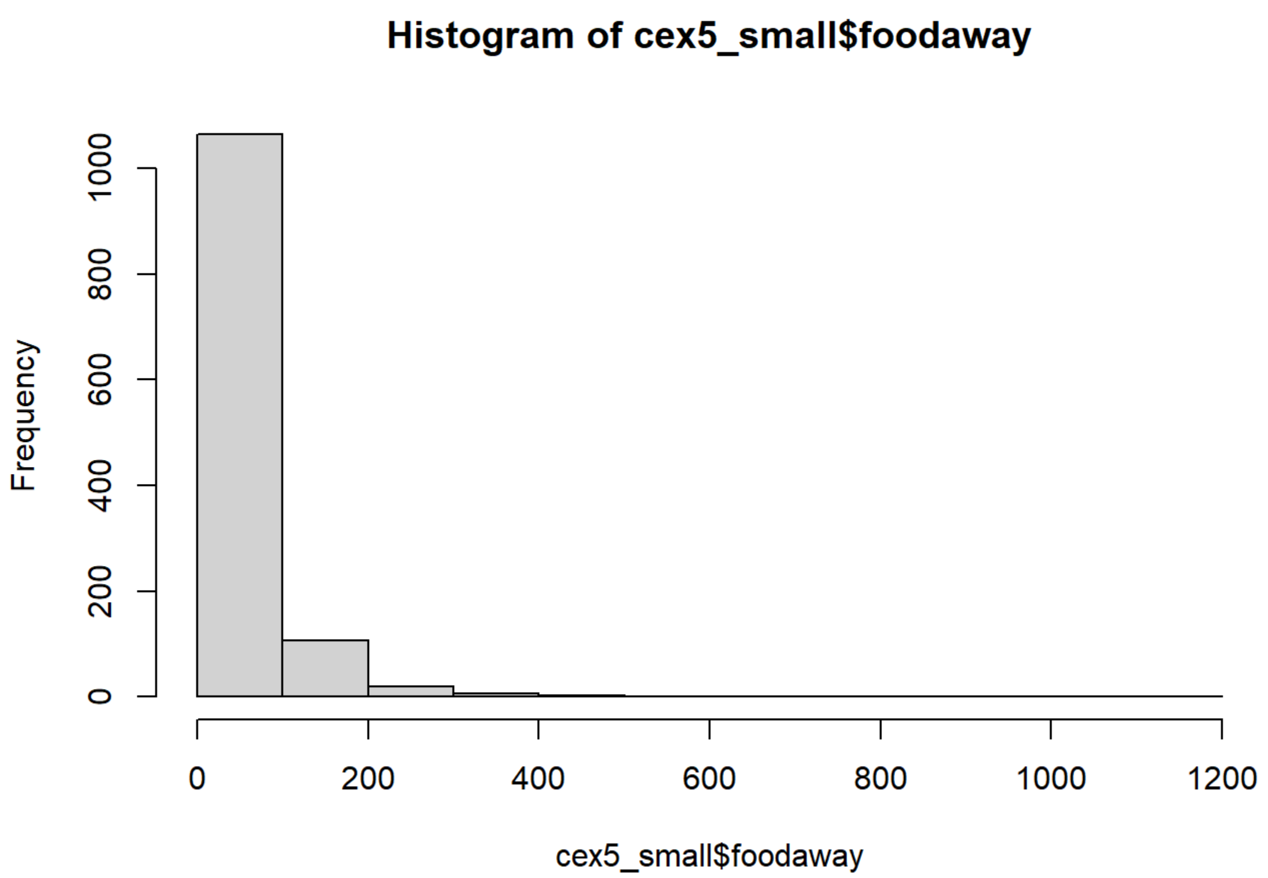
$$MATHSCORE = 483.777 + 1.435 \times AIDE$$

1. Intercept (483.777): The estimated average math score for students in regular-sized classes without an aide is 483.777.
2. Slope for aide (1.435): Having a teacher aide increases the math score by 1.435 points on average. With a p-value of 0.664, this effect is also not statistically significant.

Conclusion: Both p-values (0.161 and 0.664) are highly insignificant, failing to reject the null hypothesis. Therefore, we cannot conclude that adding a teacher aide has impact on either reading or math scores.

2.25(a)

```
library(readxl)
cex5_small <- read_excel("C:/finance ecomatrics/data/cex5_small.xlsx")
hist(cex5_small$foodaway)
```




```
summary(cex5_small$foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   12.04   32.55   49.27   67.50 1179.00
```

mean =49.27 median =32.55 25th percentile =12.04 75th percentile =67.50

2.25(b)

```
d1 <- subset(cex5_small, advanced == 1)
mean(d1$foodaway)
```

```
## [1] 73.15494
```

```
median(d1$foodaway)
```

```
## [1] 48.15
```

```
d2 <- subset(cex5_small, college == 1)
mean(d2$foodaway)
```

```
## [1] 48.59718
```

```
median(d2$foodaway)
```

```
## [1] 36.11
```

```
d3 <- subset(cex5_small, advanced == 0 & college == 0)
mean(d3$foodaway)
```

```
## [1] 39.01017
```

```
median(d3$foodaway)
```

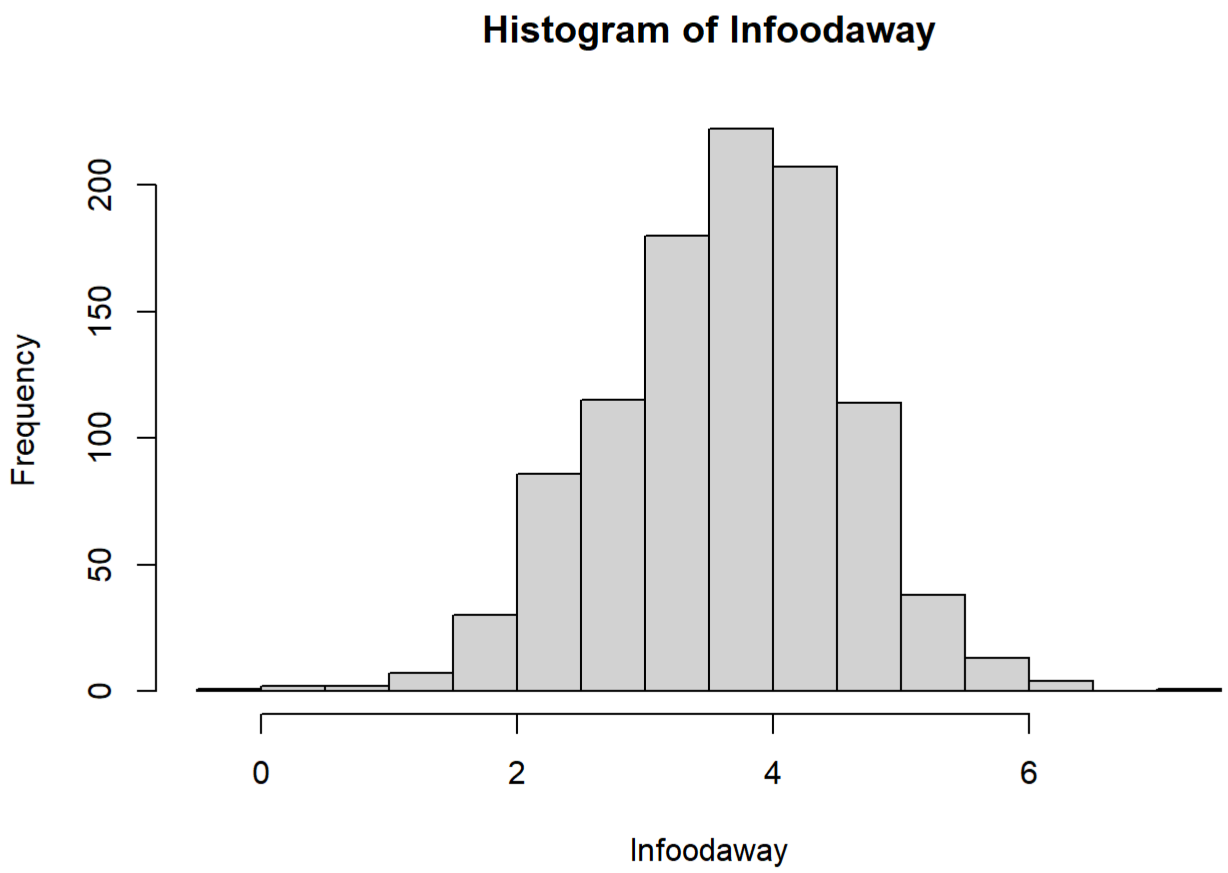
```
## [1] 26.02
```

2.25(c)

```
lnfoodaway <- log(cex5_small$foodaway)
lnfoodaway[is.infinite(lnfoodaway)] <- NA
sum(is.na(lnfoodaway))
```

```
## [1] 178
```

```
hist(lnfoodaway)
```




```
summary(lnfoodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
## -0.3011  3.0759  3.6865  3.6508  4.2797  7.0724     178
```

There are 178 fewer values of ln(FOODAWAY). Observations with zero expenditure are dropped because the log transformation is undefined at zero.So ln(foodaway) has less observations.

2.25(d)

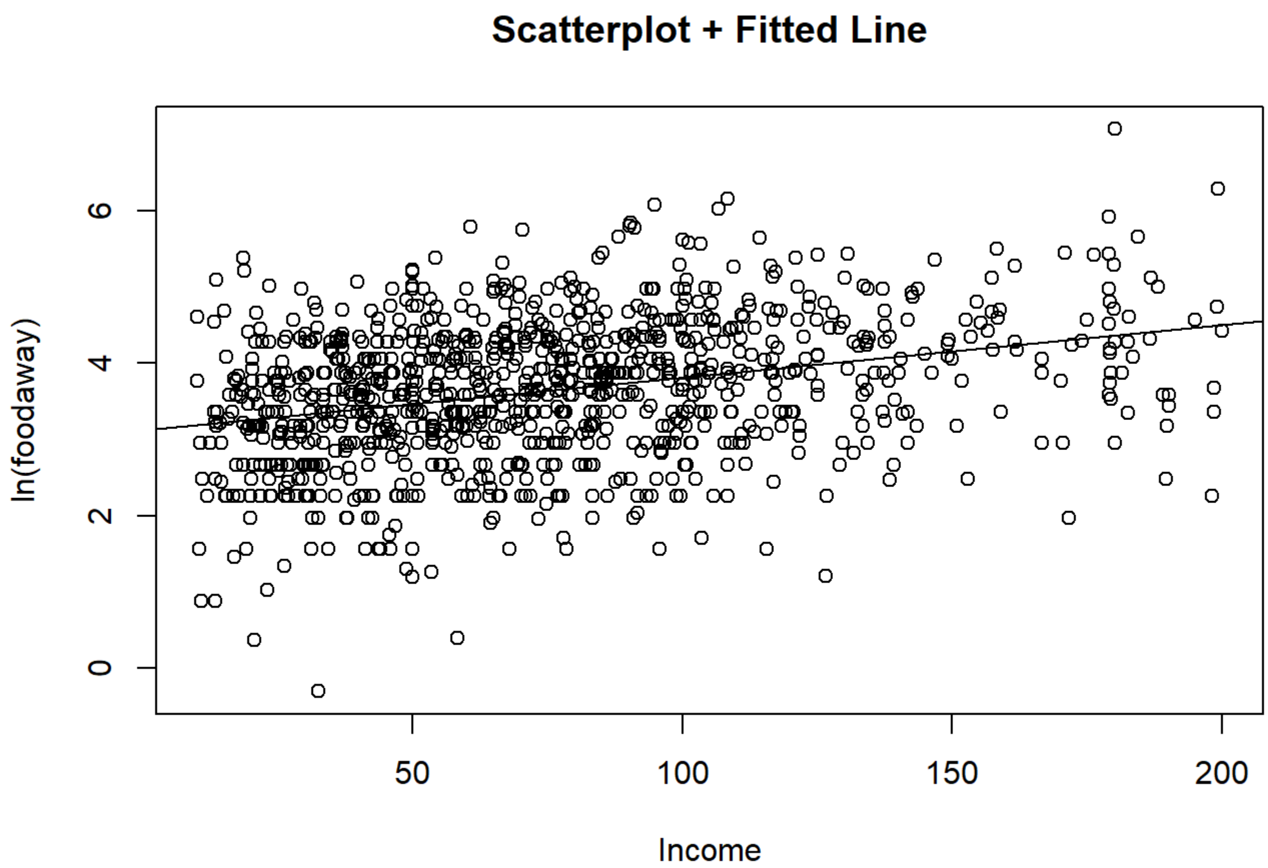
```
cex5_small$lnfoodaway <- log(cex5_small$foodaway)
cex5_small$lnfoodaway[is.infinite(cex5_small$lnfoodaway)] <- NA
regmod <- lm(lnfoodaway ~ income, data = cex5_small, , na.action = na.exclude)
summary(regmod)
```

```
##
## Call:
## lm(formula = lnfoodaway ~ income, data = cex5_small, na.action = na.exclude)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
## income      0.0069017  0.0006546   10.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
##   (因為不存在，178 個觀察量被刪除了)
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

each additional $100 household income increases food away expenditures per person of about 0.69%, other factors held constant.

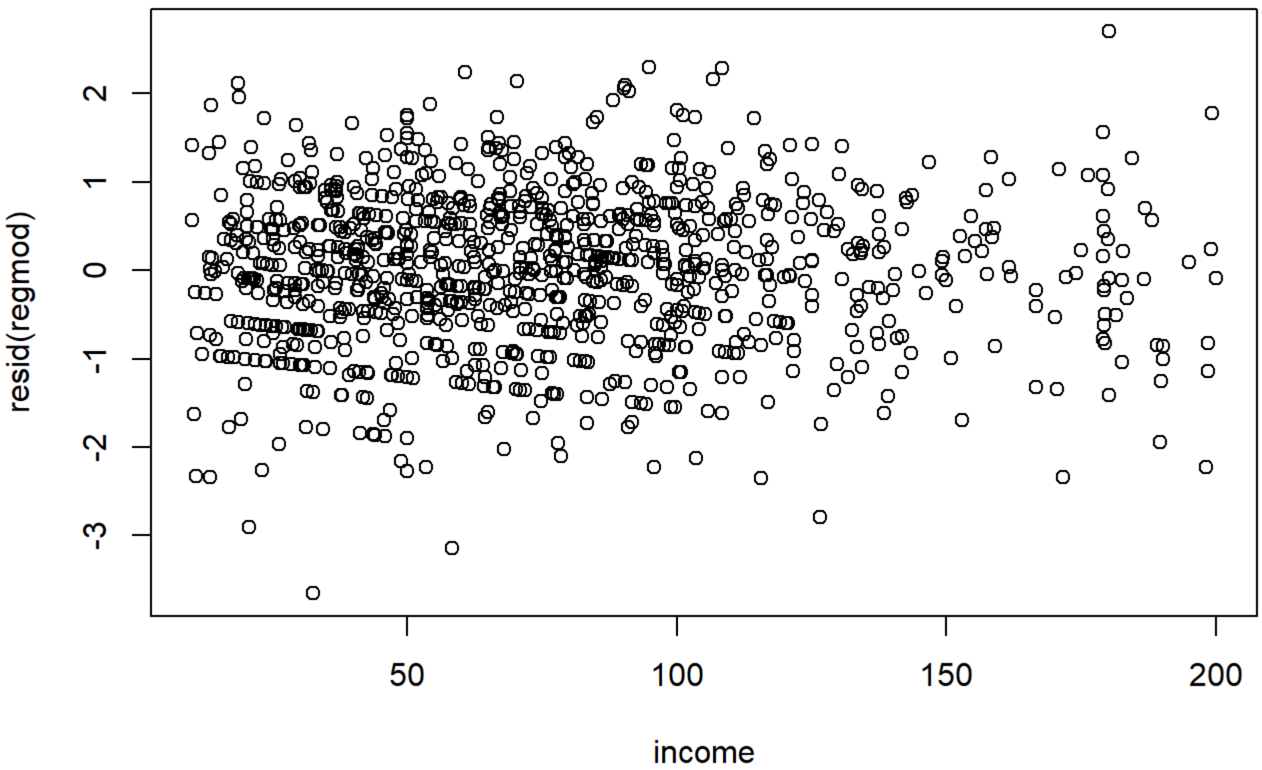
2.25(e)

```
plot(cex5_small$income, cex5_small$lnfoodaway, xlab = "Income",ylab = "ln(foodaway)",main =
"Scatterplot + Fitted Line")
abline(regmod)
```



2.25(f)

```
plot(resid(regmod) ~income, data = cex5_small)
```



The OLS residuals do appear randomly distributed with no obvious patterns. There are fewer observations at higher incomes, so there is more white space.